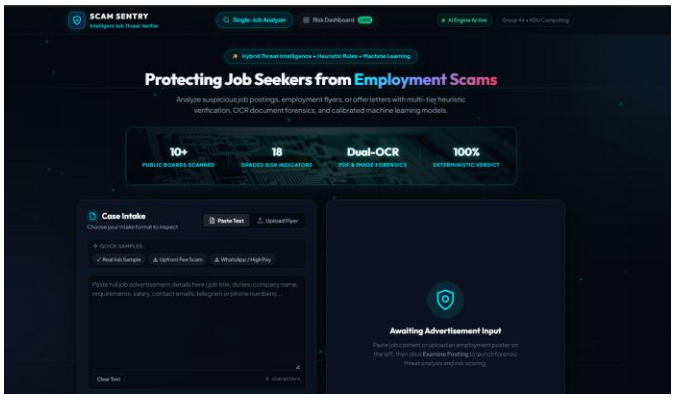
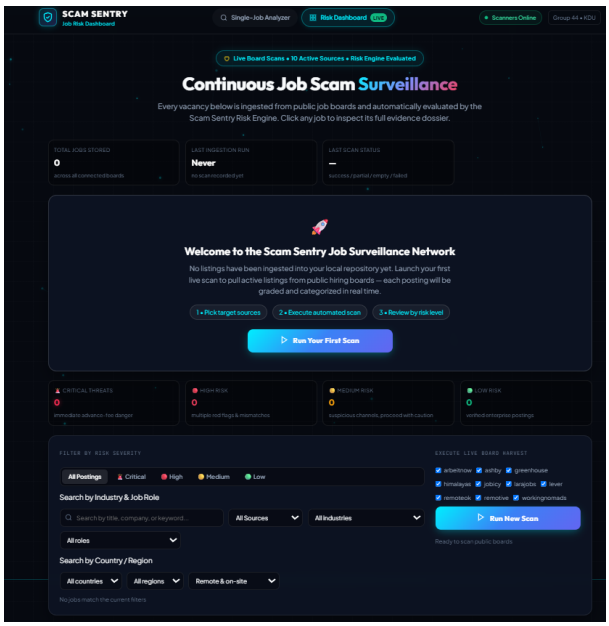
ESSENTIALS OF ARTIFICIAL INTELLIGENCE

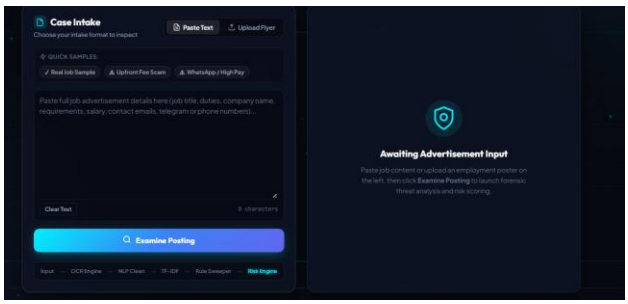
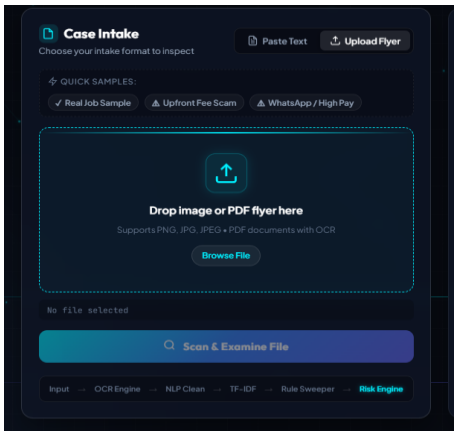
INDIVIDUAL CONTRIBUTION SHEET

Student Name	**PDH Rangana**	**Registration No.**	**D/COE/25/0025**
Degree / Intake	**Computer engineering Intake 42**	**Group No.**	**44**
Project Title	**Scam Sentry: An Intelligent Fake Job Advertisement Detection System**		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
1. Cyber-Defense UI/UX Design & Visual Ergonomics	Designed and developed the modern dark cyber-defense visual identity with glassmorphism styling and responsive layout. Created custom interactive visual telemetry including an HTML5 Neural Constellation Canvas background, dynamic laser radar sweep animation, and an SVG circular risk dial gauge for intuitive threat comprehension.	HTML5 Canvas API; CSS3 Glassmorphism; CSS Custom Properties (Design Tokens); JavaScript (ES6+); Visual Ergonomics	fakejob_app/static/style.css fakejob_app/static/script.js fakejob_app/templates/index.html Interactive Risk Score Gauge & Constellation Canvas 

			
2. Frontend-Backend API Integration & Test Presets	Integrated the frontend interface with the backend Flask API (/api/analyze/text, /api/analyze/file). Implemented quick-test preset buttons (one-click sample job postings) so users can test the system instantly, and handled asynchronous request submission, loading spinners, and dynamic verdict rendering without page reloads.	JavaScript (Fetch API, Async/Await), DOM Manipulation, Flask REST Endpoints	fakejob_app/static/script.js Quick test preset buttons /api/analyze/text Dynamic verdict card display
2. Dark Glassmorphism UI & Drag-and-Drop File Upload	Designed and styled the responsive user interface using a	HTML5, CSS3 (Glassmorphism & Flexbox/Grid),	fakejob_app/templates/index.html fakejob_app/static/style.css Interactive upload dropzone & Circular Risk Dial

	dark cyber-defense glassmorphism theme. Implemented the interactive drag-and-drop file upload zone with hover/drag states for uploading job descriptions and PDF files, and styled the circular SVG risk score dial meter and evidence cards.	SVG, Drag & Drop API	 
4. AI Model Testing, Evaluation & Feature Mapping	Evaluated and tested the AI scam detection model using curated legitimate and fraudulent sample job postings. Analyzed how key scam indicators (such as urgency keywords, vague role descriptions, suspicious contact emails, and salary anomalies) affect the model's confidence score, and mapped the output predictions into clear risk categories (Low, Medium, High).	AI Model Testing, Feature Impact Evaluation, Gradient Boosting Model Validation, Pattern Recognition	Test sample job cases job_scam_model.pkl inference validation Risk level classification outputs (Low/Med/High)
5. Version Control, Test	Maintained	Git, GitHub Actions (CI/CD),	

Alignment & CI/CD Compliance	structured version control on GitHub (design branch). Aligned all templates, styles, and client scripts to satisfy strict automated test contracts, ensuring 100% compliance across 364 automated tests including CSP security headers and UI rendering specifications.	Pytest, Content Security Policy (CSP)	GitHub Repo: https://github.com/yasithnirmal88/scam-sentry (branch: design) 364 passing tests (test_dashboard_ux.py, test_dashboard.py)

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:-  **Date:** - 10/18/2026